



Commonwealth of Virginia

VIRGINIA DEPARTMENT OF ENVIRONMENTAL QUALITY

www.deq.virginia.gov

David L. Bulova
Secretary of Natural and Historic Resources

Michael S. Rolband, PE, PWD, PWS Emeritus
Director

April 27, 2026

Mr. Christopher Sweet
Environmental Health & Safety Director
Cashvad Ventures LLC
1212 New York Avenue NW, Suite 1000
Washington, DC 20005

Location: Prince William County
Registration No.: 74118

Dear Mr. Sweet:

Attached is a permit approval to construct and operate a data center in accordance with the provisions of the Virginia State Air Pollution Control Board Regulations for the Control and Abatement of Air Pollution.

In the course of evaluating the application and arriving at a final decision to approve the project, the Department of Environmental Quality (DEQ) deemed the application complete on April 23, 2026.

This permit contains legally enforceable conditions. Failure to comply may result in a Notice of Violation and/or civil charges. Please read all permit conditions carefully.

This permit approval to construct and operate shall not relieve Cashvad Ventures LLC of the responsibility to comply with all other local, state, and federal permit regulations.

The proposed diesel fired non-emergency engine generator sets (engine gen-sets) may be subject to 40 CFR 63, Maximum Achievable Control Technology, (MACT) Subpart ZZZZ and 40 CFR 60, New Source Performance Standard (NSPS), Subpart IIII. Virginia has not accepted delegation of these rules. In summary, the units may be required to comply with certain federal emission standards and operating limitations. The Department of Environmental Quality (DEQ) advises you to review the referenced MACT and NSPS to ensure compliance with applicable emission and operational limitations. As the owner/operator you may be also responsible for any monitoring, notification, reporting and recordkeeping requirements of the MACT and NSPS. Notifications shall only be sent to EPA, Region III.

To review any federal rules referenced in the above paragraph or in the attached permit, the US Government Publishing Office maintains the text of these rules at www.ecfr.gov, Title 40, Part 60 and 63.

The Board's Regulations as contained in Title 9 of the Virginia Administrative Code 5-170-200 provide that you may request a formal hearing from this case decision by filing a petition with the Board within 30 days after this case decision notice was mailed or delivered to you. Please consult the relevant regulations for additional requirements for such requests.

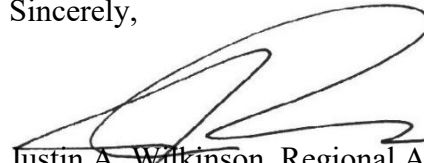
As provided by Rule 2A:2 of the Supreme Court of Virginia, you have 30 days from the date you actually received this permit or the date on which it was mailed to you, whichever occurred first, within which to initiate an appeal of this decision by filing a Notice of Appeal with:

Michael S. Rolband, Director
Department of Environmental Quality
P. O. Box 1105
Richmond, VA 23218

If this permit was delivered to you by mail, three days are added to the thirty-day period in which to file an appeal. Please refer to Part Two A of the Rules of the Supreme Court of Virginia for information on the required content of the Notice of Appeal and for additional requirements governing appeals from decisions of administrative agencies.

If you have any questions concerning this permit, please contact Ms. Katie DeVoss at (571) 866-6090 or katie.devoss@deq.virginia.gov.

Sincerely,



Justin A. Wilkinson, Regional Air Permit Manager
Virginia Department of Environmental Quality
justin.wilkinson@deq.virginia.gov
Northern Regional Office
13901 Crown Court, Woodbridge, VA 22193
(703) 583-3800

JAW/KD/74118 mNSR (2026-04-27)

Attachment: Permit



Commonwealth of Virginia

VIRGINIA DEPARTMENT OF ENVIRONMENTAL QUALITY

www.deq.virginia.gov

David L. Bulova
Secretary of Natural and Historic Resources

Michael S. Rolband, PE, PWD, PWS Emeritus
Director

STATIONARY SOURCE PERMIT TO CONSTRUCT AND OPERATE

In compliance with the Federal Clean Air Act and the Commonwealth of Virginia Regulations for the Control and Abatement of Air Pollution,

Cashvad Ventures LLC
1212 New York Avenue NW, Suite 1000
Washington, DC 20005
Registration No.: 74118

is authorized to construct and operate

non-emergency engine generator-sets (gen-sets)
cooling towers

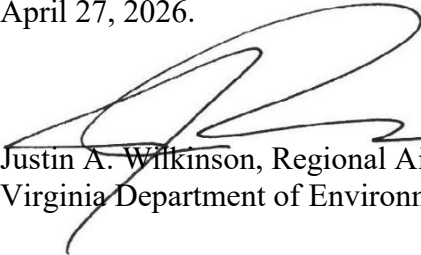
located at

10400 Harry J Parrish Blvd
Manassas, VA 20110
(Prince William County)

in accordance with the Conditions of this permit.

Approved on

April 27, 2026.


Justin A. Wilkinson, Regional Air Permit Manager
Virginia Department of Environmental Quality

Permit consists of 22 pages.
Permit Conditions 1 to 35.

INTRODUCTION

This permit document is based on the following permit approval and the respective permit application:

- mNSR permit dated April 27, 2026, based on the permit application dated October 22, 2025, including supplemental information dated December 19, 2025, January 5, 2026, February 10, 2026, February 20, 2026, March 9, 2026, March 16, 2026, March 18, 2026, April 1, 2026, and April 23, 2026.

Any changes in the permit application specifications or any existing facilities which alter the impact of the facility on air quality may require a permit. Failure to obtain such a permit prior to construction may result in enforcement action. In addition, this facility may be subject to additional applicable requirements not listed in this permit.

Words or terms used in this permit shall have meanings as provided in 9VAC5-10-10 of the State Air Pollution Control Board Regulations for the Control and Abatement of Air Pollution. The regulatory reference or authority for each condition is listed in parentheses () after each condition. The enabling permit program, or permit programs is provided below each permit condition in the regulatory authority parenthetical as follows: 9VAC5-80-850 for Article 5, 9VAC5-80-1180 for Article 6, 9VAC5-80-1985 for Article 8, and 9VAC5-80-2050 for Article 9. The most recent effective date for a term or condition is listed in brackets []. When identical conditions for one or more emission units are combined, the effective date listed in this permit does not alter the prior effective date(s) for any such conditions as issued in a previous permit action. In accordance with 9VAC5-80-1120F, any condition not marked as state-only enforceable (SOE) is state and federally enforceable.

Annual requirements to fulfill legal obligations to maintain current stationary source emissions data will necessitate a prompt response by the permittee to requests by the Department of Environmental Quality (DEQ) for information to include, as appropriate: process and production data; changes in control equipment; and operating schedules. Such requests for information from the DEQ will either be in writing or by personal contact.

The availability of information submitted to the DEQ will be governed by applicable provisions of the Freedom of Information Act, §§ 2.2-3700 through 2.2-3714 of the Code of Virginia, § 10.1-1314 (addressing information provided to the Board) of the Code of Virginia, and 9VAC5-170-60 of the State Air Pollution Control Board Regulations. Information provided to federal officials is subject to appropriate federal law and regulations governing confidentiality of such information.

Equipment List - Equipment at this facility covered by this permit consists of:

Equipment to be Constructed:

Ref. No(s).	Equipment Description	Maximum Rated Capacity (each unit)	Add-on Controls	Referenced Federal Requirements	Original Permit Date
EG-GR-1 through EG-GR-30	(30) Caterpillar C175-20 non-emergency engine gen-sets	4,000 ekW 5,647 bhp	Selective Catalytic Reduction (SCR) and catalyzed Diesel Particulate Filter (cDPF)	N/A	April 27, 2026
EG-GR-31 through EG-GR-195	(165) Caterpillar 3516E non-emergency engine gen-sets	3,000 ekW 4,393 bhp	SCR and cDPF	N/A	April 27, 2026
CT1 through CT10	(10) Cooling Towers	1,715 gpm	N/A	N/A	April 27, 2026

Specifications included in the above table is for informational purposes only and do not form enforceable terms or conditions of the permit.

PROCESS REQUIREMENTS

1. **Emission Controls (SCR)** – Nitrogen oxide (NO_x) emissions from the non-emergency diesel engine gen-sets (Ref. Nos. EG-GR-1 through EG-GR-195) shall be controlled by closed loop Selective Catalytic Reduction (SCR). Each SCR system shall be equipped with a temperature probe to continuously monitor the catalyst bed exhaust temperature while the engine gen-set is operational. Engine exhaust gas shall be treated with urea, except for periods of start-up, shutdown, or malfunction, when the catalyst bed exhaust temperature of 572°F is achieved.

The permittee shall operate the engine gen-set and SCR such that the catalyst bed exhaust temperature does not exceed 977°F. The SCR shall be provided with adequate access for inspection and shall be in operation when the engine gen-sets are operating as stated above. (9VAC5-80-1180 and 9VAC5-50-260) [4/27/2026]

2. **Emission Controls (cDPF)** – Particulate (PM, PM₁₀, and PM_{2.5}), carbon monoxide (CO), and volatile organic compounds (VOC) emissions from the non-emergency diesel engine gen-sets (Ref. Nos. EG-GR-1 through EG-GR-195) shall be controlled by a catalyzed Diesel Particulate Filter (cDPF). The cDPF shall be provided with adequate access for inspection and shall be in operation when the non-emergency diesel engine gen-set is operating. The cDPF shall be equipped with a backpressure monitor that notifies the permittee when the high backpressure limit of the engine is approached. (9VAC5-80-1180 and 9VAC5-50-260) [4/27/2026]
3. **Emission Controls (Cooling Towers)** – Particulate (PM, PM₁₀, and PM_{2.5}) from the cooling towers (Ref. Nos. CT1 through CT10) shall be controlled by drift eliminator designed for 0.001% loss. The drift eliminator shall be provided with adequate access for inspection and shall be in operation when the cooling tower is operating. (9VAC5-80-1180 and 9VAC5-50-260) [4/27/2026]
4. **Monitoring Devices (SCR and cDPF)** – Each non-emergency diesel engine gen-set (Ref. Nos. EG-GR-1 through EG-GR-195) shall be equipped with devices to continuously measure and record the following:
 - a. The SCR catalyst bed exhaust temperature. The information shall be recorded at a minimum frequency of once every fifteen minutes, and correlated to run date, engine operating hours, and fuel consumption.
 - b. The NO_x emissions concentration measured after the SCR catalyst, expressed in ppm. The information shall be recorded at a minimum frequency of once every fifteen minutes, and correlated to run date, engine operating hours, and fuel consumption.
 - c. cDPF catalyst bed temperature at a minimum frequency of once every fifteen minutes during the operation of each engine gen-set. The information shall be correlated to run date and fuel consumption.

- d. The differential pressure drop across the cDPF. The filter shall be observed by the permittee with a frequency as recommended by the manufacturer.

Each monitoring device shall be installed, maintained, calibrated and operated in accordance with approved procedures which shall include, as a minimum, the manufacturer's written requirements or recommendations. Each monitoring device shall be provided with adequate access for inspection and shall be in operation when the SCR and cDPF are operating.

(9VAC5-80-1180 D) [4/27/2026]

5. **Monitoring**

- a. Fuel Flow: Each non-emergency diesel engine gen-set (Ref. Nos. EG-GR-1 through EG-GR-195) shall be equipped with a device to continuously measure and record individual fuel consumption (in gallons) for each engine gen-set.
- b. Engine Operating Hours: Each non-emergency diesel engine gen-set (Ref. Nos. EG-GR-1 through EG-GR-195) shall be equipped with a non-resettable hour meter which measures the duration of time that each engine gen-set is operated.

Each monitoring device (as required in a. and b. above) shall be observed by the permittee with a frequency of not less than once each day the non-emergency diesel engine gen-set is operated. The permittee shall keep a log of these observations.

Each monitoring device shall be installed, maintained, calibrated (as appropriate), and operated in accordance with approved procedures which shall include, as a minimum, the manufacturer's written requirements or recommendations. The details of the monitoring device calibrations are to be arranged with the Regional Air Compliance Manager of the DEQ's Northern Regional Office (NRO).

Each monitoring device shall be provided with adequate access for inspection and shall be in operation when the engine gen-sets are operating.

(9VAC5-80-1180 D) [4/27/2026]

OPERATING LIMITATIONS

- 6. **Operation of the Engine Gen-Sets** – The permittee shall operate and maintain each non-emergency diesel engine gen-set (Ref. Nos. EG-GR-1 through EG-GR-195) and control device according to the manufacturer's written instructions or procedures developed by the permittee that are approved by the engine manufacturer. In addition, the permittee may only change those settings that are permitted by the manufacturer and do not increase air emissions.

(9VAC5-80-1180) [4/27/2026]

7. **Operating Limitations (Ozone Season)** – No non-emergency diesel engine gen-set (Ref. Nos. EG-GR-1 through EG-GR-195) shall be operated for scheduled maintenance checks and readiness testing (Scheduled MCRT), stack testing, or operational training (that involves fuel combustion) between the hours of 7 a.m. to 5 p.m. any day during May 1 through September 30. The permittee may petition the Regional Air Compliance Manager of DEQ's NRO, for exceptions to this requirement, with approvals made on a case-by-case basis.
(9VAC5-80-1180) [4/27/2026]
8. **Operating Limitations (Ozone Season) – Integration Operational Period** – During the integration operational period of each non-emergency diesel engine gen-set (Ref. Nos. EG-GR-1 through EG-GR-195), any operation of the unit (that involves fuel combustion) between the hours of 7 a.m. to 5 p.m. any day during the ozone season of May 1 through September 30 shall only occur if the forecast Air Quality Index (AQI) for ozone as published on the AirNow website (www.airnow.gov/https://airnow.gov/) for Northern Virginia for that day is less than or equal to 100. In the event that AirNow-EnviroFlash (www.enviroflash.info) issues an Air Alert for Northern Virginia/DC area for a day which the forecasted AQI for ozone was less than or equal to 100, operation of each unit (which involves fuel combustion) shall be minimized to the maximum extent practical. The integration operational period is defined as the period of time beginning with the first time the affected unit is started on-site and ending when the affected unit is fully integrated with the source's electrical system.
(9VAC5-80-1180) [4/27/2026]
9. **Fuel Specification** – The approved fuel for the non-emergency diesel engine gen-sets (Ref. Nos. EG-GR-1 through EG-GR-195) is diesel fuel oil. For the purposes of this permit document, diesel fuel oil is defined as ultra-low sulfur diesel fuel oil (ULSD), renewable diesel (RD), or a blend of these fuels, and shall meet the specifications below:

DIESEL FUEL OIL:

- a. Does not exceed the American Society for Testing and Materials (ASTM) specification, D975, for ultra-low sulfur grade 1-D or 2-D S15; or
- b. Has a maximum sulfur content not to exceed 0.0015% by weight (15 ppm), and either a minimum cetane number of forty (40) or maximum aromatic content of thirty-five percent (35%) by volume.

Exceedance of these specifications may be considered credible evidence of an exceedance of emission limits. A change in the fuel type or the fuel sulfur content may require a permit to modify and operate.

(9VAC5-80-1180 and 9VAC5-50-260) [4/27/2026]

10. **Fuel Certification** – The permittee shall obtain a certification from the fuel supplier with each shipment of diesel fuel oil. Each fuel supplier certification shall include the following:
- a. The name of the fuel supplier;
 - b. The date on which the diesel fuel was received;
 - c. The quantity of diesel fuel oil delivered in the shipment; and
 - d. A statement that the fuel complies with the requirements of Condition 9, Fuel Specifications.:
 - i. complies with the ASTM specifications for Grade No. 1-D S15 or Grade No. 2-D S15 or
 - ii. has a sulfur content per shipment not to exceed 0.0015% by weight (15 ppm) and either a minimum cetane number of forty or maximum aromatic content of thirty-five percent by volume.

Alternatively, the permittee must obtain approval from DEQ's NRO Regional Air Compliance Manager, if other documentation will be used to certify the diesel fuel type.

Fuel sampling and analysis, independent of that used for certification, as may be periodically required or conducted by DEQ may be used to determine compliance with the fuel specifications stipulated in Condition 9. Exceedance of these specifications may be considered credible evidence of the exceedance of emission limits.
(9VAC5-80-1180) [4/27/2026]

11. **Diesel Fuel Throughput Limits** –

- a. The non-emergency diesel engine gen-sets (Ref. Nos. EG-GR-1 through EG-GR-195) combined shall consume no more than 3,745,979 gallons of diesel fuel oil per year for all operations, calculated daily as the sum of each consecutive 365-day period.
- b. The non-emergency engine gen-sets (Ref. Nos. EG-GR-1 through EG-GR-195) when emissions are not controlled by SCR or cDPF, combined, shall consume no more than 293,003 gallons of diesel fuel oil per year, calculated daily as the sum of each consecutive 365-day period.

Compliance for each consecutive 365-day period shall be demonstrated daily by adding the total for the most recently completed calendar day to the individual daily totals for the preceding 364 days.
(9VAC5-80-1180) [4/27/2026]

12. **Cooling Tower Limits** – The Total Dissolved Solid (TDS) concentration in the process water in each cooling tower (Ref. Nos. CT1 through CT10) shall not exceed 2,500 ppm. (9VAC5-80-1180) [4/27/2026]

EMISSION LIMITS

13. **Emission Limits (Hourly - Engines)** – Emissions from the operation of each non-emergency diesel engine gen-set (Ref. Nos. EG-GR-1 through EG-GR-195) shall not exceed the limits specified below:

Pollutant	Caterpillar C175-20 (Ref. Nos. EG-GR-1 through EG-GR-30)	Caterpillar C175-20 (Ref. Nos. EG-GR-1 through EG-GR-30)	Caterpillar 3516E (Ref. Nos. EG-GR-31 through EG-GR-195)	Caterpillar 3516E (Ref. Nos. EG-GR-31 through EG-GR-195)
	Controlled*	Uncontrolled	Controlled*	Uncontrolled
Nitrogen Oxides (NO _x as NO ₂)	6.74 lbs/hr	84.77 lbs/hr	4.74 lbs/hr	59.56 lbs/hr
Carbon Monoxide (CO)	2.02 lbs/hr	20.44 lbs/hr	0.97 lbs/hr	9.81 lbs/hr
Volatile Organic Compounds (VOC)	0.18 lbs/hr	1.82 lbs/hr	0.06 lbs/hr	0.58 lbs/hr
Particulate Matter (PM)	0.07 lbs/hr	0.75 lbs/hr	0.18 lbs/hr	1.78 lbs/hr
Particulate Matter (PM ₁₀)	0.36 lbs/hr	1.03 lbs/hr	0.40 lbs/hr	2.00 lbs/hr
Particulate Matter (PM _{2.5})	0.36 lbs/hr	1.03 lbs/hr	0.40 lbs/hr	2.00 lbs/hr

* Controlled by SCR, cDPF

Compliance with these emission limits shall be based on the proper operation and maintenance of the non-emergency diesel engine gen-sets and pollution control devices, or by testing, if required.
(9VAC5-80-1180 and 9VAC5-50-260) [4/27/2026]

14. **Emission Limits (Annual - Engines)** – Emissions from the operation of the non-emergency diesel engine gen-sets (Ref. Nos. EG-GR-1 through EG-GR-195) shall not exceed the limits specified below:

Pollutant	Uncontrolled Operations	All Operations
Nitrogen Oxides (NO _x as NO ₂)	51.04 tpy	98.90 tpy
Carbon Monoxide (CO)	18.67 tpy	40.43 tpy
Volatile Organic Compounds (VOC)	2.81 tpy	6.12 tpy
Particulate Matter (PM)	1.24 tpy	2.71 tpy
Particulate Matter (PM ₁₀)	1.40 tpy	4.69 tpy
Particulate Matter (PM _{2.5})	1.40 tpy	4.69 tpy

These emissions are derived from the estimated overall emission contribution from operating limits. Exceedance of the operating limits may be considered credible evidence of the exceedance of emission limits. Compliance with the annual emission limits may be determined as stated in Conditions 1, 2, 9, 11, and 13.
(9VAC5-80-1180) [4/27/2026]

15. **Emission Limits (Cooling Towers)** – Emissions from the operation of the cooling towers (Ref. Nos. CT1 through CT10) shall not exceed the limits specified below:

Pollutant	Cooling Towers Combined
Particulate Matter (PM)	0.94 tpy
Particulate Matter (PM ₁₀)	0.52 tpy

These emissions are derived from the estimated overall emission contribution from operating limits. Exceedance of the operating limits may be considered credible evidence of the exceedance of emission limits. Compliance with the annual emission limits may be determined as stated in Conditions 3 and 12.
(9VAC5-80-1180 and 9VAC5-50-260) [4/27/2026]

16. **Visible Emission Limit** – Visible emissions from each non-emergency engine gen-set (Ref. Nos. EG-GR-1 through EG-GR-195) shall not exceed 5% opacity as determined by EPA Method 9 (reference 40 CFR 60, Appendix A). This requirement applies at all times except during startup, shutdown, and malfunction.

During startup and shutdown, visible emissions from each non-emergency diesel engine gen-set shall not exceed 10% opacity except during one (1) six-minute period in any one-hour in which visible emissions shall not exceed 20% opacity as determined by EPA Method 9 (reference 40 CFR 60, Appendix A).
(9VAC5-80-1180 and 9VAC5-50-260) [4/27/2026]

INITIAL COMPLIANCE DETERMINATION

17. **Stack Test (Caterpillar C175-20)** – Initial performance tests shall be conducted on six (6) Caterpillar C175-20 non-emergency diesel engine gen-sets (Ref. Nos. EG-GR-1 through EG-GR-30) for NO_x (as NO₂) and CO using appropriate EPA reference methods as approved by the Regional Air Compliance Manager of the DEQ's NRO to determine compliance with the controlled emission limits contained in Condition 13.
 - a. Emissions testing of each pollutant for each selected non-emergency diesel engine gen-set shall consist of three (3) one-hour test runs under load. The average of the three (3) runs shall be reported as the short-term emission rate for that non-emergency diesel engine gen-set.
 - b. Testing shall be performed on the non-emergency diesel engine gen-sets to demonstrate compliance with the controlled NO_x and controlled CO emission limits specified in Condition 13. Testing shall be conducted with the non-emergency diesel engine gen-set operating at ≥ 90 percent of its rated capacity, unless multiple load band testing is approved by DEQ;
 - c. Recorded non-emergency diesel engine gen-set operational information shall include, but not be limited to:
 - i. Generator load/kilowatt output.
 - ii. Fuel consumption and fuel sulfur content of the diesel fuel oil.
 - iii. NO_x concentration after the catalyst.
 - iv. SCR catalyst bed exhaust temperature.
 - d. Perform testing to demonstrate compliance within 120 days after the integration operational period has commenced. If this deadline falls within the ozone season (May 1 through September 30) the facility shall perform testing to demonstrate compliance within 30 days after the end of the ozone season. The permittee may petition the Regional Air Compliance Manager of DEQ's NRO, for exceptions to this requirement, with approvals made on a case-by-case basis. Tests shall be conducted and reported and data reduced as set forth in 9VAC5-50-30;

- e. The details of the tests are to be arranged with the Regional Air Compliance Manager of DEQ's NRO. The permittee shall submit the test protocol to the Regional Air Compliance Manager of DEQ's NRO, at least 30 days prior to testing to ensure adequate time for DEQ approval. If the test protocol is received by the DEQ with less than 30 days for review and acceptance, DEQ approval may not be issued in a timely manner to allow for testing to take place according to the permittee's schedule;
- f. Should conditions occur which would require rescheduling the testing, the permittee shall notify the Regional Air Compliance Manager of DEQ's NRO, in writing, within seven (7) days of the scheduled test date or as soon as the rescheduling is deemed necessary; and
- g. Two (2) copies (one (1) paper copy and one (1) electronic copy) of the test results shall be submitted to the Regional Air Compliance Manager, DEQ's NRO within 60 days after test completion and shall conform to the test report format enclosed with this permit.

(9VAC5-50-30 and 9VAC5-80-1200) [4/27/2026]

18. **Stack Test (Caterpillar 3516E)** – Initial performance tests shall be conducted twenty-two (22) Caterpillar 3516E non-emergency diesel engine gen-sets (Ref. Nos. EG-GR-31 through EG-GR-195) for NO_x (as NO₂) and CO using appropriate EPA reference methods as approved by the Regional Air Compliance Manager of the DEQ's NRO to determine compliance with the controlled emission limits contained in Condition 13.

- a. Emissions testing of each pollutant for each selected non-emergency diesel engine gen-set shall consist of three (3) one-hour test runs under load. The average of the three (3) runs shall be reported as the short-term emission rate for that non-emergency diesel engine gen-set.
- b. Testing shall be performed on the non-emergency diesel engine gen-sets to demonstrate compliance with the controlled NO_x and controlled CO emission limits specified in Condition 13. Testing shall be conducted with the non-emergency diesel engine gen-set operating at ≥ 90 percent of its rated capacity, unless multiple load band testing is approved by DEQ;
- c. Recorded non-emergency diesel engine gen-set operational information shall include, but not be limited to:
 - i. Generator load/kilowatt output.
 - ii. Fuel consumption and fuel sulfur content of the diesel fuel oil.
 - iii. NO_x concentration after the catalyst.

- iv. SCR catalyst bed exhaust temperature.
- d. Perform testing to demonstrate compliance within 120 days after the integration operational period has commenced. If this deadline falls within the ozone season (May 1 through September 30) the facility shall perform testing to demonstrate compliance within 30 days after the end of the ozone season. The permittee may petition the Regional Air Compliance Manager of DEQ's NRO, for exceptions to this requirement, with approvals made on a case-by-case basis. Tests shall be conducted and reported and data reduced as set forth in 9VAC5-50-30;
- e. The details of the tests are to be arranged with the Regional Air Compliance Manager of DEQ's NRO. The permittee shall submit the test protocol to the Regional Air Compliance Manager of DEQ's NRO, at least 30 days prior to testing to ensure adequate time for DEQ approval. If the test protocol is received by the DEQ with less than 30 days for review and acceptance, DEQ approval may not be issued in a timely manner to allow for testing to take place according to the permittee's schedule;
- f. Should conditions occur which would require rescheduling the testing, the permittee shall notify the Regional Air Compliance Manager of DEQ's NRO, in writing, within seven (7) days of the scheduled test date or as soon as the rescheduling is deemed necessary; and
- g. Two (2) copies (one (1) paper copy and one (1) electronic copy) of the test results shall be submitted to the Regional Air Compliance Manager, DEQ's NRO within 60 days after test completion and shall conform to the test report format enclosed with this permit.

(9VAC5-50-30 and 9VAC5-80-1200) [4/27/2026]

- 19. **Stack Test (Renewable Diesel)** – Initial performance tests shall be conducted on two (2) non-emergency diesel engine gen-sets (Ref. Nos. EG-GR-1 through EG-GR-195) while utilizing renewable diesel or a blend of renewable diesel and ULSD for NO_x (as NO₂) and CO using appropriate EPA reference methods as approved by the Regional Air Compliance Manager of the DEQ's NRO to determine compliance with the controlled emission limits contained in Condition 13.
 - a. Emissions testing of each pollutant for each selected non-emergency diesel engine gen-set shall consist of three (3) one-hour test runs under load. The average of the three (3) runs shall be reported as the short-term emission rate for that non-emergency diesel engine gen-set.
 - b. Testing shall be performed on the exhaust stack of the non-emergency diesel engine gen-sets to demonstrate compliance with the controlled NO_x and controlled CO emission limits specified in Condition 13. Testing shall be conducted with the non-

emergency diesel engine gen-set operating at ≥ 90 percent of its rated capacity, unless multiple load band testing is approved by DEQ;

- c. Recorded non-emergency diesel engine gen-set operational information shall include, but not be limited to:
 - i. Generator load/kilowatt output.
 - ii. Fuel consumption and fuel sulfur content of the diesel fuel oil.
 - iii. NO_x concentration after the catalyst.
 - iv. SCR catalyst bed exhaust temperature.
 - v. Differential pressure drop across the cDPF.
- d. For any non-emergency engine gen-set that has not commenced the integration operational period upon receipt of the first shipment of renewable diesel or a blend of renewable diesel and ULSD utilized by the units: Perform testing to demonstrate compliance within 120 days after the integration operational period has commenced. If this deadline falls within the ozone season (May 1 through September 30), the facility shall perform testing to demonstrate compliance within 30 days after the end of the ozone season. Tests shall be conducted and reported and data reduced as set forth in 9VAC5-50-30;
- e. For any non-emergency engine gen-set that has commenced the integration operational period upon receipt of the first shipment of renewable diesel or a blend of renewable diesel and ULSD utilized by the units: Perform testing to demonstrate compliance within 120 days of the renewable diesel fuel or a blend of renewable diesel and ULSD first utilized by the affected units. The permittee may petition the DEQ's NRO Air Compliance Manager for an extension to this deadline, with approvals made on a case-by-case basis. If the applicable deadline falls within the ozone season (May 1 through September 30), the facility shall perform testing to demonstrate compliance within 30 days after the end of the ozone season. Tests shall be conducted and reported and data reduced as set forth in 9VAC5-50-30;
- f. The details of the tests are to be arranged with the Regional Air Compliance Manager of DEQ's NRO. The permittee shall submit the test protocol to the Regional Air Compliance Manager of DEQ's NRO, at least 30 days prior to testing to ensure adequate time for DEQ approval. If the test protocol is received by the DEQ with less than 30 days for review and acceptance, DEQ approval may not be issued in a timely manner to allow for testing to take place according to the permittee's schedule;
- g. Should conditions occur which would require rescheduling the testing, the permittee shall notify the Regional Air Compliance Manager of DEQ's NRO, in writing, within

seven (7) days of the scheduled test date or as soon as the rescheduling is deemed necessary; and

- h. Two (2) copies (one (1) paper copy and one (1) electronic copy) of the test results shall be submitted to the Regional Air Compliance Manager, DEQ's NRO within 60 days after test completion and shall conform to the test report format enclosed with this permit.

(9VAC5-50-30 and 9VAC5-80-1200) [4/27/2026]

- 20. **Visible Emissions Evaluation** – Concurrent with the initial performance tests required in Conditions 17, 18, and 19, Visible Emission Evaluations (VEE) in accordance with 40 CFR Part 60, Appendix A, Method 9, shall also be conducted by the permittee on the non-emergency diesel engine gen-sets selected for initial performance testing. The details of the tests are to be arranged with the Regional Air Compliance Manager of DEQ's NRO. The permittee shall submit a VEE protocol in conjunction with the initial stack test protocol required by Conditions 17, 18, and 19 at least 30 days prior to testing.

- a. Should conditions prevent concurrent opacity observations, the Regional Air Compliance Manager of the DEQ's NRO shall be notified in writing, within seven (7) days, and visible emissions testing shall be rescheduled within 30 days. Rescheduled testing shall be conducted under the same operating conditions as the initial performance tests.
- b. Two (2) copies of the test result (one (1) hard copy and one (1) electronic copy) shall be submitted to the Regional Air Compliance Manager of the DEQ's NRO within 60 days after test completion and shall conform to the test report format enclosed with this permit (Attachment A).

(9VAC5-50-30 and 9VAC5-80-1200) [4/27/2026]

CONTINUING COMPLIANCE DETERMINATION

- 21. **Cooling Towers** – Once per calendar year, the permittee shall sample and analyze the process water from one cooling tower (Ref. Nos. CT1 through CT10) to demonstrate compliance with the TDS limit in Condition 12, utilizing EPA approved procedures or equivalent methods as determined by the Air Compliance Manager of DEQ's NRO.

(9VAC5-80-1180) [4/27/2026]

22. **Facility Construction** – The non-emergency diesel engine gen-sets (Ref. Nos. EG-GR-1 through EG-GR-195) shall be constructed so as to allow for emissions testing upon reasonable notice at any times, using appropriate methods. This includes constructing the facility/equipment such that volumetric flow rates and pollutant emission rates can be determined by applicable test methods and providing a stack or duct that is free from cyclonic flow. Sampling ports shall be provided when requested at the appropriate locations and safe sampling platforms and access shall be provided.
(9VAC5-50-30 F and 9VAC5-80-1180) [4/27/2026]
23. **Emission Testing/Visible Emissions Evaluation** – Upon request by the DEQ, the permittee shall conduct stack tests and/or VEEs of the non-emergency diesel engine gen-sets (Ref. Nos. EG-GR-1 through EG-GR-195) to demonstrate compliance with the emission limits contained in this permit. The details of the tests shall be arranged with the Regional Air Compliance Manager of the DEQ's NRO.
(9VAC5-80-1200 and 9VAC5-50-30 G) [4/27/2026]

RECORDS

24. **On Site Records** – The permittee shall maintain records of emission data and operating parameters as necessary to demonstrate compliance with this permit. The content and format of such records shall be arranged with the Regional Air Compliance Manager of the DEQ's NRO. These records shall include, but are not limited to:
- a. Documentation from the manufacturer that each non-emergency diesel engine gen-set is certified to meet the EPA Tier 2 emission standards.
 - b. Engine information including make, model, serial number, model year, maximum engine power (bhp), and engine displacement for each non-emergency diesel engine gen-set.
 - c. Documentation from the manufacturer that the cooling towers (Ref. Nos. CT1 through CT10) meet the design specifications required by Condition 3.
 - d. Operation and control device monitoring records for each non-emergency engine gen-set equipped with a SCR (Ref. Nos. EG-GR-1 through EG-GR-195) as required in Conditions 4.a and 4.b. This includes records of the SCR catalyst exhaust bed temperature and NO_x emission concentration as measured by the SCR continuous monitoring device.
 - e. Operation and control device monitoring records for each non-emergency engine gen-set equipped with a cDPF (Ref. Nos. EG-GR-1 through EG-GR-195) as required in Conditions 4.c and 4.d. This includes records of the cDPF catalyst bed temperature and differential pressure drop.
 - f. A log of monitoring device observations as required by Condition 5.

- g. The manufacturer's written operating instructions or procedures developed by the owner/operator that are approved by the engine manufacturer for each non-emergency diesel engine gen-set.
- h. Records, as necessary, to demonstrate compliance with the operating limitations of Condition 7; which includes but is not limited to: times, dates and reasons for operation of each diesel engine gen-set that was operating between May 1 and September 30.
- i. To verify compliance with Condition 8, maintain records of:
 - i. The forecasted AQI, as determined by the AirNow website for Northern Virginia, for ozone for the days that a non-emergency diesel engine gen-set operated during the integration operational period;
 - ii. The measured AQI, as determined by the AirNow website for Northern Virginia, for ozone for the days that a non-emergency diesel engine gen-set operated during the integration operational period;
 - iii. Documentation recording any Air Alerts issued for that operating day, as determined by AirNow-EnviroFlash; and
 - iv. Details of commissioning activities, to include, but not limited to, clock hours, and duration.
- j. Daily and annual fuel consumption of each non-emergency diesel engine gen-set (Ref. Nos. EG-GR-1 through EG-GR-195), for all purposes, with the annual fuel consumption calculated daily as the sum of each consecutive 365-day period.
- k. Daily and annual fuel consumption for the combined operation of the non-emergency diesel engine gen-sets (Ref. Nos. EG-GR-1 through EG-GR-195) for all operations, calculated daily as the sum of each consecutive 365-day period, to verify compliance with the fuel throughput limitations specified in Condition 11.a.
- l. Daily and annual fuel consumption for the combined operation of the non-emergency diesel engine gen-sets (Ref. Nos. EG-GR-1 through EG-GR-195) when emissions are not controlled by SCR or cDPF, calculated daily as the sum of each consecutive 365-day period, to verify compliance with the fuel throughput limitations specified in Condition 11.b.
- m. Records of TDS sampling and analytic results used to determine compliance with the cooling tower process water TDS limit in Condition 12.

- n. Daily and annual emissions calculations for NO_x (as NO₂), CO, VOC, PM, PM₁₀, and PM_{2.5} from the combined operation of the non-emergency diesel engine gen-sets (Ref. Nos. EG-GR-1 through EG-GR-195), when emissions are uncontrolled by SCR or cDPF, with annual emissions calculated daily, as the sum of each consecutive 365-day period, to verify compliance with the annual emission limits in Condition 14.
- o. Daily and annual emissions calculations for NO_x (as NO₂), CO, VOC, PM, PM₁₀, and PM_{2.5} from the combined operation of the non-emergency diesel engine gen-sets (Ref. Nos. EG-GR-1 through EG-GR-195), for all operations, with annual emissions calculated daily, as the sum of each consecutive 365-day period, to verify compliance with the annual emission limits in Condition 14.
- p. Annual emissions calculations for PM and PM₁₀ from the combined operation of the cooling towers (Ref. Nos. CT1 through CT10), with annual emissions calculated monthly, as the sum of each consecutive 11-month period, to verify compliance with the annual emission limits in Condition 15.
- q. All fuel supplier certifications.
- r. Results of all stack tests and VEEs.
- s. Records of maintenance, operator training, and records as required by Condition 30.
- t. Records of changes in settings that are permitted by the manufacturer of the non-emergency diesel engine gen-sets.

Compliance for the consecutive 12-month period in the subsections above (as applicable) shall be demonstrated monthly by adding the total for the most recently completed month to the individual monthly totals for the preceding 11 months.

Compliance for the consecutive 365-day period in the subsections above (as applicable) shall be demonstrated daily by adding the total for the most recently completed day to the individual daily totals for the preceding 364 days.

These records shall be available for inspection by the DEQ and shall be current for the most recent five (5) years.

(9VAC5-80-1180 and 9VAC5-50-50) [4/27/2026]

NOTIFICATIONS

25. **Initial Notifications** – The permittee shall furnish written notification of the items below to the Regional Air Compliance Manager of the DEQ’s NRO at the following address:

Regional Air Compliance Manager
Department of Environmental Quality
13901 Crown Court
Woodbridge, VA 22193

The permittee shall submit notification(s) for each building containing the information as described below:

- a. The actual date on which installation of the non-emergency diesel engine gen-sets (Ref. Nos. EG-GR-1 through EG-GR-195) and cooling towers (Ref. Nos. CT1 through CT10) commenced in each building, within thirty (30) days after such date. The notification must contain the following:
 - i. Name and address of the permittee;
 - ii. The building;
 - iii. Unit reference number of the initial unit installed; and
 - iv. The date installation commenced.
- b. The start and end dates of the integration operational period for each non-emergency diesel engine gen-set (Ref. Nos. EG-GR-1 through EG-GR-195) within fifteen (15) days after the last engine gen-set at each building completes its integration operational period. If a period of construction is paused or halted for ≥ 45 days, this notification shall be provided to the DEQ within fifteen (15) days after completion of the integration operational period for the most recently installed engine gen-set. The notification must contain the following:
 - i. Unit reference number;
 - ii. Engine information including make, model, engine family, serial number, model year, maximum engine power, engine displacement, fuel used;
 - iii. Installation date; and
 - iv. Integration operational period start and end dates.
- c. The date of commencing of construction of the cooling towers (Ref. Nos. CT1 through CT10) within 30 days after such date.

- d. The actual date of initial startup of the cooling towers (Ref. Nos. CT1 through CT10) within 15 days after the last cooling tower at each building completes initial startup.

For the purpose of this notification, the integration operational period is defined as the period of time beginning with the first time the affected unit is started on-site and ending when the affected unit is fully integrated with the source's electrical system.
(9VAC5-50-20) [4/27/2026]

- 26. **Renewable Diesel Notification** –Upon receipt of the first shipment of renewable diesel or a blend of renewable diesel and ULSD, the permittee shall furnish written notification of the items below to the Air Compliance Manager of the DEQ's NRO at the address in Condition 25.

- a. The actual date on which the shipment was received within fifteen (15) days after such date. The notification must include the following:
 - i. Name and address of the permittee;
 - ii. The address of the affected source;
 - iii. Engine gen-sets (with reference numbers) utilizing the fuel in the shipment; and
 - iv. Fuel certification (as provided in Condition 10).

(9VAC5-80-1180) [4/27/2026]

GENERAL CONDITIONS

- 27. **Permit Invalidation** – This permit to construct the non-emergency diesel engine gen-sets (Ref. Nos. EG-GR-1 through EG-GR-195) and the cooling towers (Ref. Nos. CT1 through CT10) shall become invalid, unless an extension is granted by the DEQ, if:

- a. A program of continuous construction is not commenced within 18 months from the 'Original Permit Date' specified in the equipment list in the Introduction section of this permit; or if
- b. A program of construction is discontinued for a period of 18 months or more, or is not completed within a reasonable time.

(9VAC5-80-1210)

28. **Permit Suspension/Revocation** – This permit may be suspended or revoked if the permittee:
- a. Knowingly makes material misstatements in the permit application or any amendments to it;
 - b. Fails to comply with the conditions of this permit;
 - c. Fails to comply with any emission standards applicable to a permitted emissions unit;
 - d. Causes emissions from the stationary source which result in violations of, or interfere with the attainment and maintenance of, any ambient air quality standard; or
 - e. Fails to operate in conformance with any applicable control strategy, including any emission standards or emissions limitations, in the State Implementation Plan in effect at the time an application for this permit is submitted.

(9VAC5-80-1210 G)

29. **Right of Entry** – The permittee shall allow authorized local, state, and federal representatives, upon the presentation of credentials:
- a. To enter upon the permittee's premises on which the facility is located or in which any records are required to be kept under the terms and conditions of this permit;
 - b. To have access to and copy at reasonable times any records required to be kept under the terms and conditions of this permit or the State Air Pollution Control Board Regulations;
 - c. To inspect at reasonable times any facility, equipment, or process subject to the terms and conditions of this permit or the State Air Pollution Control Board Regulations; and
 - d. To sample or test at reasonable times.

For purposes of this condition, the time for inspection shall be deemed reasonable during regular business hours or whenever the facility is in operation. Nothing contained herein shall make an inspection time unreasonable during an emergency.

(9VAC5-170-130 and 9VAC5-80-1180)

30. **Maintenance/Operating Procedures** – At all times, including periods of start-up, shutdown, and malfunction, the permittee shall, to the extent practicable, maintain and operate the affected source, including associated air pollution control equipment, in a manner consistent with good air pollution control practices for minimizing emissions.

The permittee shall take the following measures in order to minimize the duration and frequency of excess emissions:

- a. Develop a maintenance schedule and maintain records of all scheduled and non-scheduled maintenance.
- b. Maintain an inventory of spare parts.
- c. Have available written operating procedures for equipment. These procedures shall be based on the manufacturer's recommendations, at a minimum.
- d. Train operators in the proper operation of all such equipment and familiarize the operators with the written operating procedures, prior to their first operation of such equipment. The permittee shall maintain records of the training provided including the names of trainees, the date of training and the nature of the training.

Records of maintenance and training shall be maintained on site for a period of five years and shall be made available to DEQ personnel upon request.
(9VAC5-50-20 E and 9VAC5-80-1180 D)

31. **Record of Malfunctions** – The permittee shall maintain records of the occurrence and duration of any bypass, malfunction, shutdown or failure of the facility or its associated air pollution control equipment that results in excess emissions for more than one hour. Records shall include the date, time, duration, description (emission unit, pollutant affected, cause of malfunction), corrective action, preventive measures taken and name of person generating the record.
(9VAC5-20-180 J and 9VAC5-80-1180 D)

32. **Notification for Facility or Control Equipment Malfunction** – The permittee shall furnish notification to the Regional Air Compliance Manager of the DEQ's NRO of malfunctions of the affected facility or related air pollution control equipment that may cause excess emissions for more than one hour. Such notification shall be made no later than four daytime business hours after the malfunction is discovered. The permittee shall provide a written statement giving all pertinent facts, including the estimated duration of the breakdown, within 14 days of discovery of the malfunction. When the condition causing the failure or malfunction has been corrected and the equipment is again in operation, the permittee shall notify the Regional Air Compliance Manager of the DEQ's NRO.
(9VAC5-20-180 C and 9VAC5-80-1180)

33. **Violation of Ambient Air Quality Standard** – The permittee shall, upon request of the DEQ, reduce the level of operation or shut down a facility, as necessary to avoid violating any primary ambient air quality standard and shall not return to normal operation until such time as the ambient air quality standard will not be violated.
(9VAC5-20-180 I and 9VAC5-80-1180)
34. **Change of Ownership** – In the case of a transfer of ownership of a stationary source, the new owner shall abide by any current minor NSR permit issued to the previous owner. The new owner shall notify the NRO of the change of ownership within 30 days of the transfer.
(9VAC5-80-1240)
35. **Permit Copy** – The permittee shall keep a copy of this permit on the premises of the facility to which it applies.
(9VAC5-80-1180)

SOURCE TESTING REPORT FORMAT

Report Cover

1. Plant name and location
2. Units tested at source (indicate Ref. No. used by source in permit or registration)
3. Test Dates.
4. Tester; name, address and report date

Certification

1. Signed by team leader/certified observer (include certification date)
2. Signed by responsible company official
3. *Signed by reviewer

Copy of approved test protocol

Summary

1. Reason for testing
2. Test dates
3. Identification of unit tested & the maximum rated capacity
4. *For each emission unit, a table showing:
 - a. Operating rate
 - b. Test Methods
 - c. Pollutants tested
 - d. Test results for each run and the run average
 - e. Pollutant standard or limit
5. Summarized process and control equipment data for each run and the average, as required by the test protocol
6. A statement that test was conducted in accordance with the test protocol or identification & discussion of deviations, including the likely impact on results
7. Any other important information

Source Operation

1. Description of process and control devices
2. Process and control equipment flow diagram
3. Sampling port location and dimensioned cross section Attached protocol includes: sketch of stack (elevation view) showing sampling port locations, upstream and downstream flow disturbances and their distances from ports; and a sketch of stack (plan view) showing sampling ports, ducts entering the stack and stack diameter or dimensions

Test Results

1. Detailed test results for each run
2. *Sample calculations
3. *Description of collected samples, to include audits when applicable

Appendix

1. *Raw production data
2. *Raw field data
3. *Laboratory reports
4. *Chain of custody records for lab samples
5. *Calibration procedures and results
6. Project participants and titles
7. Observers' names (industry and agency)
8. Related correspondence
9. Standard procedures

* Not applicable to visible emission evaluations